

Axiom 2025/26 - I Got One More In Me
 Basics

Definition: Random Definitions

- $\mathbb{N} = \mathbb{N}_0$, ie $\mathbb{N} = \{n \in \mathbb{Z} : n \geq 0\}$
- In this course we only use binary codes, ie $F = \mathbb{F}_2$, the two-element field
- For $a, b \in \mathbb{Z}$, a divides b , or $a|b$, if $\exists c \in \mathbb{Z}$ s.t. $b = ac$
- $a, b \in \mathbb{Z}$ are **relatively prime** or **coprime** if $\gcd(a, b) = 1$

Theorem: Random Theorems

- Well-Ordering:** All sets of positive ints $\neq \emptyset$ contain a least element

Definition: Primitive Pythagorean Triples

A **Pythagorean Triple** is a set of numbers $(a, b, c) \in \mathbb{N}$ such that $a^2 + b^2 = c^2$. Note if (a, b, c) is a PT, so is (da, db, dc) . A **Primitive Pythagorean Triple** is a PT such that a, b, c have no common factor. Such a, b, c have one of a, b odd and the other even. *Square of an odd is odd, square of an even is even. If both even then a, b, c divisible by 2. If both odd work through the eqn with $a = 2k + 1, b = 2n + 1$ to see all divisible by 4. Indeed, every PPT with (a, b, c) with a odd*

$$\text{has, for } s \geq t \geq 1 \text{ odd coprime integers, } a = st, b = \frac{s^2 - t^2}{2}, c = \frac{s^2 + t^2}{2}.$$

Theorem: The Division Algorithm

- Let $a, b \in \mathbb{Z}$ with $b > 0$. Then $\exists$ unique $q, r \in \mathbb{Z}$ s.t. $a = qb + r$ and $0 \leq r < b$. q is the **quotient** and r is the **remainder**.
- If $ax + by = c$ then $\gcd(a, b) = d$ divides c since $d|a$ and $d|b$
- Euclid's Lemma:** If $a, b, p \in \mathbb{Z}$, p prime, and $p|ab$; then $p|a$ or $p|b$.

Lemma: Bézout's Lemma

Let a, b be nonzero integers with $\gcd(a, b) = d$. Then $\exists s, t \in \mathbb{Z}$ s.t. $as + bt = d$. Moreover any common divisor of a and b divides d .

Theorem: Fermat's Last Theorem

For $n, x, y, z \in \mathbb{Z}$, and $n > 2$, there are no integer solutions to $x^n + y^n = z^n$.

Theorem: The Extended Euclidean Algorithm

- Let $a, b \in \mathbb{Z}$ be s.t. $a \geq b > 0$. Then $\gcd(a, b)$ can be found as follows:
 - By Division Algorithm $\exists$ unique q_1, r_1 s.t. $a = q_1 b + r_1$ with $0 \leq r_1 < b$
 - If $r_1 > 0$, apply the Division Algorithm to b, r_1 for $b = q_2 r_1 + r_2$.
 - If $r_2 > 0$, apply the DA to r_1, r_2 for $r_1 = q_3 r_2 + r_3$
 - Repeat *ad infinitum*, the last non-zero remainder is $\gcd(a, b)$

Example: Finding GCD By Euclidean Algorithm

Find $\gcd(112, 27)$.
 $112 = 4(27) + 4$ $27 = 6(4) + 3$ $4 = 1(3) + 1$ $3 = 3(1) + 0$
 So $\gcd(112, 27) = 1$ and 112, 27 are coprime

Definition: Congruence

- a is **congruent** to b mod n , written $a \equiv b \pmod{n}$, if $n|(a - b)$, ie $\exists q \in \mathbb{Z}$ s.t. $a = nq + b$. For $a, b, c, d, n \in \mathbb{Z}$ with $n > 0$, $a \equiv b \pmod{n}$, $c \equiv d \pmod{n}$:
 - $a + c \equiv b + d \pmod{n}$ $ac \equiv bd \pmod{n}$
 - $a - c \equiv b - d \pmod{n}$

Definition: System Of Residues

A **complete system of residues modulo** n is a set $I \subseteq \mathbb{Z}$ s.t. every $k \in \mathbb{Z}$ is congruent mod n to exactly one $i \in I$

Theorem: Wilson's Theorem

If p is prime then $(p - 1)! \equiv -1 \pmod{p}$

Major Theorem: The Chinese Remainder Theorem

Let $m_1, m_2, \dots, m_k \in \mathbb{N}$ be **pairwise coprime**. Then given $a_1, a_2, \dots, a_k \in \mathbb{Z}$, the system of congruences

$$\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ x &\equiv a_2 \pmod{m_2} \\ &\vdots \\ x &\equiv a_k \pmod{m_k} \end{aligned}$$

has a unique soln mod N ($N = m_1 \cdot m_2 \cdot \dots \cdot m_k$). Constructive proof:
Proof. Let $N_i = n/m_i$ be the product of all m_j with $j \neq i$. Since m_i is coprime to m_j for $i \neq j$, we must have $\gcd(N_i, m_i) = 1$. Then since N_i and m_i are coprime, N_i must have an inverse mod m_i , ie $\exists x_i \in \mathbb{Z}$ s.t. $N_i x_i \equiv 1 \pmod{m_i}$ (to find x_i use the Euclidean algorithm to find numbers s_i . $N_i x_i m_i y_i = 1$). Then consider the product $a_i N_i x_i$, which is s.t. $a_i N_i x_i \equiv 0 \pmod{m_j}$ for $i \neq j$, and $a_i N_i x_i \equiv a_i \pmod{m_i}$. Thus $x = \sum_{i=1}^k a_i N_i x_i$ precisely satisfies $x \equiv a_i \pmod{m_i}$ for all i . $\square$

Example: Applying The CRT

Solve the system of congruences

$$\begin{aligned} x &\equiv 1 \pmod{2} \\ x &\equiv 3 \pmod{5} \\ x &\equiv 4 \pmod{7} \end{aligned}$$

We have $N = 2 \cdot 5 \cdot 7 = 70$, $N_1 = 5 \cdot 7 = 35$, $N_2 = 2 \cdot 7 = 14$, $N_3 = 2 \cdot 5 = 10$. Calculate by Euclidean Algorithm or inspection for:

$$\begin{aligned} 35 \cdot 1 &\equiv 1 \pmod{2} \\ 14 \cdot 4 &\equiv 1 \pmod{5} \\ 10 \cdot 5 &\equiv 1 \pmod{7} \end{aligned}$$

so we have $x_1 = 1, x_2 = 4, x_3 = 5$.
 Thus $x \equiv (1 \cdot 35 \cdot 1) + (3 \cdot 14 \cdot 4) + (4 \cdot 10 \cdot 5) \equiv 403 \equiv 53 \pmod{70}$.

Theorem: Applying The CRT

Key fact - if m, n are coprime integers, and $a \in \mathbb{Z}$, the congruence $x \equiv a \pmod{m \cdot n}$ has the same integer solutions as the system

$$\begin{aligned} x &\equiv a \pmod{m} \\ x &\equiv a \pmod{n} \end{aligned}$$

Thus knowing $x \pmod{N}$ is equivalent to knowing $x \pmod{a}$ each of the primes in the prime decomposition of N , and to solve such systems we can simply decompose it to a larger system where all m_i are coprime.

Example: Is It A Quadratic Residue?

Is 7 a quadratic residue modulo 53?
 Calculate $(\frac{7}{53})$. By Rules 4 and 3, $(\frac{7}{53}) = (\frac{53}{7}) = (\frac{4}{7})$.
 By Rule 2 $(\frac{4}{7}) = (\frac{2}{7})^2 = 1$. Thus 7 is a quadratic residue modulo 53.

Example: Is It A Quadratic Residue? (weird)

Let p be prime and suppose $p = 17k + 1$ for some $k > 0 \in \mathbb{Z}$. Is 17 a quadratic residue modulo p ?
 17 is a QR modulo p iff $(\frac{17}{p}) = (\frac{17}{17k+1}) = 1$. To apply Rule 4: note $17 \equiv 1 \pmod{4}$; since $k > 0, p > 17$ so $p, 17$ are distinct; and $p, 17$ are odd as 2 is the only odd prime and $p > 2$. Thus $(\frac{17}{17k+1}) = (\frac{17k+1}{17}) = (\frac{17k+1-17k}{17}) = (\frac{1}{17}) = 1$. So 17 is a quadratic residue modulo p .

Ex.: Using Quadratic Residue To Prove Non-Divisibility

Show there does not exist $x \in \mathbb{Z}$ s.t. $x^{2022} + 9$ is divisible by 43. Prove by contradiction: assume such x exists. Then $x^{2022} + 9 \equiv 0 \pmod{43}$ so $-9 \equiv (x^{1011})^2 \pmod{43}$, thus -9 is a quadratic residue modulo 43. Now consider $(\frac{-9}{43})$. By Rule 2 $(\frac{-9}{43}) = (\frac{-1}{43})(\frac{9}{43})^2$. By Rule 1 $(\frac{-1}{43}) \equiv (-1)^{42/2} \equiv (-1)^{21} \equiv -1 \pmod{43}$. By Rule 5 $(\frac{9}{43}) = -(\frac{43}{9}) = -(\frac{1}{3}) = -1$. Thus $(\frac{-9}{43}) = -1 \cdot -1 \cdot -1 = -1$ and -9 is not a quadratic residue modulo 43, so we have a contradiction. Thus no such x exists.

Example: Random Congruence Thing

Alice finds all integer solutions $x \in \{0, 1, \dots, 2025\}$ to $x^2 \equiv 1 \pmod{77}$. Mike finds all integer solutions $x \in \{0, 1, \dots, 2025\}$ to $x^{10} \equiv 1 \pmod{77}$. Spencer finds all integer solutions $x \in \{0, 1, \dots, 2025\}$ to $x^{70} \equiv 1 \pmod{77}$. Given that all their solutions are correct, which of the following are true?:
 1 A & M find exactly the same solns
 5 M finds more solns than A
 2 A & S find exactly the same solns
 6 M & S find exactly the same solns
 3 S finds more solns than A
 7 A finds no solns
 4 A finds more solns than M
 First note all solutions found are coprime with 77. Alice finds x s.t. both 7 and 11 divide $x^2 - 1$. Mike finds x s.t. both 7 and 11 divide $x^{10} - 1$. By Fermat's Little Theorem $11|x^{10} - 1$ as x is coprime to 11. Also by Fermat's Little Theorem $7|x^6 - 1$. Thus Mike finds all x s.t. $7|x \cdot \gcd(10, 6) - 1 = x^2 - 1$. Thus Mike finds more solns than Alice since by the CRT there exist x s.t. $x^2 \equiv 1 \pmod{7}$ and $x^2 \not\equiv 1 \pmod{11}$. Spencer finds x s.t. 7 and 11 divide $x^{70} - 1$. Note 11 always divides $x^{70} - 1$ for x not divisible by 11. Again by Fermat's Little Theorem $7|x^6 - 1$ for x not divisible by 7, so these x satisfy $7|x \cdot \gcd(6, 70) - 1 = x^2 - 1$ thus Mike and Spencer find the same solutions. So options 3, 5, 6 are correct.

Example: Random Prime Congruence Modulo 5 Thing

Let $p \in \mathbb{N}$, and $p > 5$ be prime. Suppose $p|n^2 - 5$. Show that p must be congruent to either 1 or 4 modulo 5. Observe that $5|(n^2 - 5)$, so 5 is a quadratic residue modulo p , so $(\frac{5}{p}) = 1$. Suppose for a contradiction that either $p = 5k + 2$ or $p = 5k + 3$ for some integer $k \geq 0$. By Rule 4, $(\frac{5}{p}) = (\frac{p}{5})$. If $p = 5k + 2$ then $(\frac{p}{5}) = (\frac{2}{5}) = -1$; and if $p = 5k + 3$ then $(\frac{p}{5}) = (\frac{3}{5}) = -1$; so in either case we have a contradiction. *This answer seems completely terrible but it was a given past paper solution ☹*

Binary Codes

Major Definition: Linear Binary Codes

A **binary word** is a sequence of 1s and 0s. The **word length** is the number of digits in the word. A **binary code** is a set of binary words. A code C is **linear** iff for all $y, z \in C$, $x + y \in C$. Binary words which are elements of a binary code are **codewords**. We can ascribe meaning to codewords (eg $(1, 1, 1) =$ 'yes' and $(0, 0, 1) =$ 'no'). A code is a **block code** if all words have the same length.
 • A **repetition code of length** n is obtained by repeating $n - 1$ times the first single symbol. Example of length 3: $\{(1, 1, 1), (0, 0, 0)\}$.
 • A binary linear code has exactly 2^k elements.
 • Binary linear codes can be considered as linear spaces over $\mathbb{F} = \{0, 1\}$.
 • $x \cdot y := \sum_{i=1}^n x_i y_i$. $(\cdot)$ is a symmetric bilinear form on $\mathbb{F}_2^n$.
 • x and y are **orthogonal** if $x \cdot y = 0$.
 • A **parity check bit** is a bit appended to a codeword to make the sum of the digits even. All binary codewords with such bits are of form $(x_1, \dots, x_{n-1}, \sum_{i=1}^{n-1} x_i)$

Definition: The Parameters Of A Linear Code

A triple (n, k, d) gives the three params of a linear code - n , the length of codewords in the code; k , the dimension of the code $\dim_{\mathbb{F}} C$, and d the minimal distance of the code. k is equal to the number of linearly independent codewords in the code. There are no linear binary codes with params (n, k, d) where $n < k + d - 1$

Definition: Neighbourhood Of A Word

For a binary word $x = (x_1, \dots, x_n)$, with all $x_i \in \{0, 1\}$ and $r \geq 0$, the **neighbourhood of x with radius r** is the set $N_r(x) = \{y = (y_1, \dots, y_n) : y_1, \dots, y_n \in \{0, 1\}, d(x, y) \leq r\}$.
 • Note that $x \in N_r(x)$ and $|N_r(x)| = \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{r}$

Definition: Hamming Distance

For C a code over $\mathbb{F}_2$, $x = (x_1, x_2, \dots, x_n), y = (y_1, y_2, \dots, y_n) \in C$; the **Hamming distance** d on C is $d(x, y) := |\{i \in \{1, \dots, n\} : x_i \neq y_i\}|$.
 • For $c \in C$, the **weight** of c is $\text{wt}(c) := d(c, 0) = |\{i \in \{1, \dots, n\} : c_i \neq 0\}|$.
 • For C, c , binary codes of the same length, $\text{wt}(c + f) = \text{wt}(c) + \text{wt}(f) - 2\text{wt}(cf)$
 • The **minimal distance** of C is $d(C) = \min\{d(x, y) : x, y \in C, x \neq y\}$, intuitively the smallest distance between any two codewords in the code.
 • For C a linear code, the minimal distance is equal to the minimum weight among non-zero codewords in the code.
 • For binary words x, y, z of length n ; $d(x, y) + d(y, z) \geq d(x, z)$

Definition: Decision Rules

For $C \subseteq \mathbb{F}^n$ a code, a **decision rule** for C is a function $\sigma : \mathbb{F}^n \rightarrow C$ which assigns to each $z \in \mathbb{F}^n$ a codeword $c \in C$.

Definition: Bit-Error

A **bit-error** is a mistake in transmission of a codeword z in a code C , where one digit of z is flipped (either from 0 to 1, or from 1 to 0). A code is r -error-correcting if it has minimal distance $d \geq 2r + 1$.

Theorem: Minimal Distance Rule

For C a code, the **minimal distance rule** states that given a codeword $z \in C$ received through a noisy channel, the receiver should choose $\sigma(z)$ to be the codeword $c \in C$ such that $d(z, c)$ is minimised. If a sender uses a code C with minimum distance $d > 2r$ and the receiver uses the MD rule, provided no more than r bit-errors are made in transmitting any codeword, the MD rule will map every received codeword to the correct transmitted one.

Definition: A Generator Matrix Of A Code

Let C be a linear code with params (n, k, d) . Since C has $\dim_{\mathbb{F}} C = k$ we can find codewords $c_1, \dots, c_k$ which form a basis of C . Then the $k \times n$ matrix with rows $c_1, \dots, c_k$ is a **generator matrix** of C

Definition: The Dual Code

For C a code with params (n, k, d) , the **dual code** $C^\perp = \{x = (x_1, \dots, x_n) : x_1, \dots, x_n \in \mathbb{F}_2, x \cdot c = 0 \forall c \in C\}$ is the orthogonal complement of C .
 • $\dim_{\mathbb{F}} C^\perp = n - k$ since $\dim_{\mathbb{F}} C = k$.
 • A code is **self-dual** if $C = C^\perp$, for example $C = \{(0, 0), (1, 1)\}$.
 • $(C^\perp)^\perp = C$.
 • $\dim C + \dim C^\perp = n$.

Definition: Parity Check Matrices

For C a code with params (n, k, d) , a generator matrix H for $C^\perp$ (a matrix whose rows form a basis of $C^\perp$) is a **parity check matrix** for C .
 • H is a $(n - k) \times n$ matrix
 • c is a codeword in C iff $Hc^T = 0$ iff $cH^T = 0$.
 • C has minimal distance d if and only if H has d lin-dep cols and does not have $d - 1$ lin-dep cols

Definition: Perfect Codes

For C a binary code of length n , minimal distance $d = 2r + 1$ for $r \geq 0 \in \mathbb{Z}$. C is a **perfect code** iff $|C|(\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{r}) = 2^n$, or equivalently if each $c \in C$ of length n is in exactly one neighbourhood $N_r(c)$

Theorem: Adjusting Code Parameters

- Let C be a linear binary code with parameters (n, k, d) .
- If $k > 1$; $\exists$ a lin-bin code with params $(n - 1, k', d')$; $k' \geq k - 1, d' \geq d$.
- If d is odd, $\exists$ a lin-bin code with params (n, k', d') ; $k' \geq k - 1, d' > d$.
- If $d > 1$, there is a lin-bin code with params $(n, k, d - 1)$

Theorem: More Code Param Theorems

- Let $r, d \in \mathbb{Z}$ s.t. $r \geq 0, d \geq 2r + 1$, and $n \in \mathbb{N}$.
 Then there are no codes C of length n s.t. $2^n < 2^k(\binom{n}{0} + \dots + \binom{n}{r})$
- Gilbert-Varshamov: Let $r, n \in \mathbb{N}, r \geq 0$. Then $\exists$ a binary code C of length n with minimal distance $d \geq 2r + 1$ s.t. $|C|(\binom{n}{0} + \dots + \binom{n}{2r}) \geq 2^n$.

Example: Maximum Cardinality Of A Code

What is the maximal number of elements in a linear binary 1-error correcting code C of length 6? C is 1-error correcting so $d \geq 2 + 1 = 3$. If $d = 3$ it is odd so there exists a code with parameters $(6, k', \geq k - 1, d' > 3)$. Given such a code, since $k > 1$ there exists a code with parameters $(6 - 1, k' \geq k - 2, d' \geq 4)$. Since $d' \geq 4$ any non-zero codewords in this code must have weight at least 4, but then their minimal distance would be at most 2 since $n = 5$. Thus the dimension of this code, k' , is at most 1. Since $1 \geq k - 2, k \leq 3$. Thus C has parameters $(6, k \leq 3, d \geq 3)$. Observe that $k = 3$ is possible with $C = \{(111000, 110011, 101010)\}$. Thus the maximum cardinality is $2^3 = 8$ since $|C| = 2^k$.

Example: Possible Parameters

Does there exist a linear binary code with parameters $(10, 5, 6)$? No: Since $k > 1$, by repeated theorem application there would exist some code with parameters $(10 - 3, k' \geq 5 - 3, d' \geq 6) = (7, k' \geq 2, d' \geq 6)$. However, in such a code, every non-zero word has weight at least 6, but any two non-zero codewords have distance at most 2, so k cannot be larger than 1.

Example: Possible Parameters Again

Is there a linear binary code C with parameters $(9, 4, 5)$? Suppose such a C exists. Then take the subcode C' of words in C with even weight, by lecture results C' is lin-bin with parameters $(9, 3, d' \geq 6)$. By lecture result (taking all codes with first digit 0) there then exists a code E with parameters $(8, 2, d \geq 6)$. Let $e_1, e_2 \in E$ be linearly independent. By lecture result, $\text{wt}(e_1), \text{wt}(e_2) \geq d \geq 6$, then they both have at least 6 digits equal to 1. Then their distance is at most 4, and E cannot have minimal distance $d \geq 6$. Thus no such C may exist.

Ex.: Forming Generator Matrix From Parity Check Matrix

Write a generator matrix for a linear binary code C which has the following parity check matrix:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

and show that every codeword in C has even weight.
 Let $x = (x_1, \dots, x_7)$ be a binary word with $x_i \in \{0, 1\}$. By lecture result, $x \in C$ iff $Hx^T = 0$, that is $x \cdot 111111 = x \cdot 1111000 = x \cdot 1100110 = x \cdot 1010101 = 0$. Since H has 7 columns and 4 rows, we know C has parameters $(7, 3, d)$. By inspection, $\{111000, 011001, 001110\}$ is a set of 3 lin-ind codewords satisfying our dot product requirements, and thus form a basis for C . Thus a generator matrix for C is $G = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}$. Also by lecture results, a codeword which is a sum of codewords with even weight has even weight, and since all rows of G have even weight and all codewords in C may be formed by a sum of these rows, all codewords in C have even weight.

Example: Bounding Minimal Distance Above

Let C be a linear code with parameters (n, k, d) and suppose $n > 2^{n-k} + 1$. Show that $d < 3$. Note the parity-check matrix has columns of length $n - k$, so there are at most 2^{n-k} distinct columns with entries 0 or 1. Thus two columns of the parity check matrix must be equal. Let w be a word s.t. $w_i = w_j = 1$ and all other $w_k = 0$. Then $wH^T = \text{column } i + \text{column } j = 0$, thus $w \in C$. Since the code is linear, the minimal distance of C equals the minimal weight of a non-zero codeword, so d is at most 2 as $\text{wt}(w) = 2$.

Example: Checking A Code Is Linear

Is $C = \{(x, y, z) \in \mathbb{F}_2^3 : x^2 + y^2 = z^2\}$ linear? How many codewords does C have? Is $D = \{(x, y, z) \in \mathbb{F}_2^3 : xyz = 0\}$ linear? Note that in $\mathbb{F}_2 = \{0, 1\}$, we have $x^2 = x, y^2 = y, z^2 = z$, so the condition on C reduces to $x + y = z$, and clearly it is linear (easy proof: check $x + w$ componentwise for $x, w \in C$). Since there are exactly two choices for x and y , and z is determined by x and y , C has exactly $2^2 = 4$ elements. D is not linear as both 100 and 011 are codewords in D but their sum 111 is not.

Example: Code Existence From Generator Matrix, Weight

Is there a linear binary code C whose generator matrix has 7 columns and 3 rows, and such that all non-zero $c \in C$ have $\text{wt}(c) > 4$? We must prove there exists C with parameters $(7, 3, n + 5)$ for $n \geq 0$. Take $E \subseteq C$, the subset of C with codewords of even weight. By lecture theory, E is a linear subspace of C , so E is linear. Observe that for $e \in E$, $\text{wt}(e)$ is even so the minimal distance of E is even, and thus is $6 + 2j</$

Euler & Rings

Major Definition: Euler's Function

For $n > 1$, the **Euler function** $\phi(n)$ is defined to be the number of natural numbers not exceeding n which are coprime with n . $\phi(1) := 1$.

- If p is prime, $\phi(p) = p - 1$
- For p prime, $\phi(p^n) = (p-1)p^{n-1} = p^n - p^{n-1} = p^n(1 - \frac{1}{p})$
- If $p_1, \dots, p_j$ are distinct primes dividing m ; $\phi(m) = m(1 - \frac{1}{p_1}) \dots (1 - \frac{1}{p_j})$
- If m, n are coprime then $\phi(m \cdot n) = \phi(m) \cdot \phi(n)$
- In general $\phi(mn) = \phi(m)\phi(n) \cdot d \cdot \phi(d)$ where $d = \gcd(m, n)$ (not given in notes but used in past paper solns ☺)
- Note that $\phi(n^2)$ is not necessarily equal to $\phi(n)^2$
- For $n \geq 3$, $\phi(n)$ is even (not a course result! use for checking only)

Major Theorem: Euler's Theorem

For $n \in \mathbb{N}$ s.t. $n > 1$, and $a \in \mathbb{Z}$ s.t. a, n coprime, then $a^{\phi(n)} - 1 \equiv 0 \pmod n$

- **Fermat's Little Thm:** If p prime and x, p coprime; $x^{p-1} \equiv 1 \pmod p$
- If p prime and x, p coprime; $x^p \equiv x \pmod p$

Definition: Carmichael Numbers

$n \in \mathbb{Z}$ is a **Carmichael number** if $a^{n-1} \equiv 1 \pmod n$ for all a coprime to n

- If n is s.t. $a^{n-1} \equiv 1 \pmod n$ for all $0 < a < n$, then n is prime

Major Definition: Rings (Different Axioms To HALG!)

A **ring** is a set R equipped with $+, \cdot: R \times R \rightarrow R$ s.t. for all $a, b, c \in R$:
 $(R, +)$ is an abelian group
• Associativity of $\cdot$: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
• Distributivity: $a \cdot (b + c) = a \cdot b + a \cdot c$; and $(a + b) \cdot c = a \cdot c + b \cdot c$
Variants of rings, with the above axioms true:
• Commutative ring: $a \cdot b = b \cdot a$
• Unital ring: $\exists 1_R \in R$ s.t. $1_R \cdot a = a \cdot 1_R = a \forall a \in R$
• Integral domain: Ring with no zero-divisors $(x, y \in R$ s.t. $x, y \neq 0, x \cdot y = 0)$

Definition: The Ring $\mathbb{Z}/n\mathbb{Z}$ or $\mathbb{Z}_n$

The elements of $\mathbb{Z}_n$ are congruence classes $\pmod n$, ie $\mathbb{Z}_n = \{\overline{0}, \overline{1}, \dots, \overline{n-1}\}$. We define the ring addition/multiplication by, for $0 \leq a, b < n$:
 $\overline{a} + \overline{b} = \overline{c}$ iff $a + b = c \pmod n$
 $\overline{a} \cdot \overline{b} = \overline{c}$ iff $a \cdot b = c \pmod n$
 $\overline{a} \cdot \overline{b} = \overline{c}$ iff $a \cdot b = c \pmod n$

Definition: Primitive Roots

For prime $p > 0$, $i \in \mathbb{Z}$ is a **primitive root modulo p** iff $i, i^2, \dots, i^{p-1}$ are pairwise distinct $\pmod p$. Ie, $i^m \neq 1 \pmod p$ for $0 < m < p-1$
Equivalent definitions: $i \in \mathbb{Z}$ is a primitive root modulo p iff:
• $p \nmid (i^j - 1)$ for any $0 < j < p-1$ and $p \mid i^{p-1} - 1$.
• For every $m \in \mathbb{Z}$ not divisible by p , $\exists \alpha \in \mathbb{N}$ s.t. $m \equiv i^\alpha \pmod p$.

Theorem: Primitive Root Theorem

For every prime $p > 0$ there exists a primitive root modulo p .
• Let $x \in \mathbb{Z}$, $n \in \mathbb{N}$, and $\alpha, \beta \in \mathbb{N}$, $\alpha, \beta > 0$.
Suppose $x^\alpha \equiv 1 \pmod n$ and $x^\beta \equiv 1 \pmod n$. Then $x^{\gcd(\alpha, \beta)} \equiv 1 \pmod n$

Major Definition: The Legendre Symbol

Let $p > 0$ be prime and $a \in \mathbb{Z}$ s.t.
 a is not divisible by p . The **Legendre symbol of a modulo p** , denoted $(\frac{a}{p})$, is equal to 1 if $a = r^2 \pmod p$ for $r \in \mathbb{Z}$, and equal to -1 otherwise.
If $(\frac{a}{p}) = 1$ then a is a **quadratic residue modulo p** .

Theorem: Legendre Symbol Theorems

- **Rule 1:** Let p be an odd prime. Then $(\frac{a}{p}) \equiv a^{(p-1)/2} \pmod p$
- **Rule 2:** $(\frac{ab}{p}) = (\frac{a}{p})(\frac{b}{p})$
- **Rule 3:** $(\frac{a}{p}) = (\frac{-a}{p})$ and moreover if $a \equiv b \pmod p$ then $(\frac{a}{p}) = (\frac{b}{p})$
- Let $p, q > 0$ be distinct odd primes. Then:
 - **Rule 4:** If $p \equiv 1 \pmod 4$ or $q \equiv 1 \pmod 4$ then $(\frac{p}{q}) = (\frac{q}{p})$
 - **Rule 5:** If $p \equiv 3 \pmod 4$ and $q \equiv 3 \pmod 4$, then $(\frac{p}{q}) = -(\frac{q}{p})$

Lemma: The Gauss Lemma

Let p be an odd prime and $a \in \mathbb{Z}$ not divisible by p . Consider $S := \{a, 2 \cdot a, 3 \cdot a, \dots, \frac{p-1}{2} \cdot a\}$. Let S' be the set of remainders of elements of S when divided by p , with each remainder in the interval $(-\frac{p}{2}, \frac{p}{2})$. Let n be the number of elements in S' smaller than 0. Then $(\frac{a}{p}) = (-1)^n$

- For prime $p > 2$, $(\frac{2}{p}) = (-1)^{(p^2-1)/8}$

Example: Applying The Gauss Lemma

Show whether 2 is a quadratic residue modulo 13 and whether 3 is a quadratic residue modulo 13 using the Gauss Lemma.
First whether $(\frac{2}{13}) = 1$: As in the Gauss Lemma, denote $S = \{2 \cdot 2, 3 \cdot 2, 4 \cdot 2, 5 \cdot 2, 6 \cdot 2\} = \{2, 4, 6, 8, 10, 12\}$. As in the proof of the Gauss Lemma, $S' = \{2, 4, 6, -5, -3, -1\}$. There are 3 negative numbers in this set, so since 3 is odd, 2 is not a quadratic residue modulo 13. Now $(\frac{3}{13})$. Again as in the GL, $S = \{3 \cdot 2, 3 \cdot 3, 3 \cdot 4, 3 \cdot 5, 3 \cdot 6, 3\} = \{3, 6, 9, 12, 15, 18\}$ and $S' = \{3, 6, -4, -1, 2, 5\}$. There are 2 negative numbers in this set, so since 2 is even, 3 is a quadratic residue modulo 13. $S = \{3 \cdot 2, 3 \cdot 3, 3 \cdot 4, 3 \cdot 5, 3 \cdot 6, 3\}$

Example: Euler's Function Equations

Is there $m \in \mathbb{N}$ s.t. for all $n > m$, $n \in \mathbb{N}$, we have $\phi(n) + \phi(2n) = \phi(3n)$?
No: Let n be arbitrarily large.
not divisible by 2, and divisible by 3. Then $\phi(2n) = \phi(2) \cdot \phi(n) = \phi(n)$, and $\phi(3n) = 3\phi(n)$. Therefore $\phi(n) + \phi(2n) = 2\phi(n) \neq 3\phi(n) = \phi(3n)$.

Example: Expression For $\phi(n)$ Given Form Of n

Let $n > 2$ be an integer s.t. $n = 4k+1$ for some $k \in \mathbb{Z}$. Show that either $\phi(n)$ is divisible by 4 or $n = p^m$ for some prime p and $m \in \mathbb{Z}$.
Since $n = 4k+1$, n is odd. If n has only one prime divisor p , then $n = p^m$ for $m \in \mathbb{Z}$ and our statement holds.
If n has two or more distinct prime divisors, then $n = \prod_{j=1}^k p_j^{\alpha_j}$ and $\phi(n) = \prod_{j=1}^k p_j^{\alpha_j-1} (p_j - 1)$. Since each $(p_j - 1)$ term is even, and $k \geq 2$, their product is divisible by 4, thus $\phi(n)$ is divisible by 4 as required.

Example: Lecture 10 Example

Let $n > 1 \in \mathbb{Z}$. Find all n s.t. $\phi(n) = 2$.
 n has some prime decomposition $n = p_1^{\alpha_1} \dots p_m^{\alpha_m}$ where all $p_i > 1$ and are pairwise distinct. By course formula $\phi(n) = (p_1^{\alpha_1-1} - p_1^{\alpha_1-1}) \dots (p_m^{\alpha_m-1} - p_m^{\alpha_m-1})$. Note that for all $i \in 1, \dots, m$, $(p_i - 1) \mid \phi(n)$, thus to satisfy our condition, $p_i - 1$ is equal to 1 or 2, so $p_i \in \{2, 3\}$.
If n has exactly one prime divisor, ie $n = p_1^{\alpha_1}$, then $\phi(n) = p_1^{\alpha_1-1} (p_1 - 1)$, then if $p_1 = 2$, $n = 4$; and if $p_1 = 3$, $n = 3$. Now assume n has more than one prime divisor. Since all $p_i \in \{2, 3\}$, $n = 2^{\alpha_1} \cdot 3^{\alpha_2}$, and $\phi(n) = 2^{\alpha_1-1} (2-1) \cdot 3^{\alpha_2-1} (3-1)$. Then $\alpha_1 - 1 = 0$ and $\alpha_2 - 1 = 0$, so $n = 6$.

Example: Gaussian Integers

Major Definition: Gaussian Integers

The **Gaussian integers** are the elements of $\mathbb{Z}[i] := \{a + ib : a, b \in \mathbb{Z}\}$. $\mathbb{Z}[i]$ is closed under $+/ -$ thus is a ring.

- For $w, z \in \mathbb{Z}[i]$, $|z| = |a + bi| = \sqrt{a^2 + b^2}$ and $|z \cdot w| = |z| \cdot |w|$
- $z \in \mathbb{Z}[i]$ is a **unit** if $\exists z' = a' + ib' \in \mathbb{Z}[i]$ s.t. $z \cdot z' = 1$. For all units $z \in \mathbb{Z}[i]$, $|z| = 1$ and $z \in \{\pm 1, \pm i\}$
- For $z, z' \in \mathbb{Z}[i]$, $z \neq 0$, we say z **divides** z' , written $z \mid z'$, iff $z' = z \cdot w$ for some $w \in \mathbb{Z}[i]$
- Let $z \in \mathbb{Z}[i]$ be non-zero and not a unit. z is an **irreducible Gaussian integer** if whenever $z = u \cdot v$ for $u, v \in \mathbb{Z}[i]$, either u or v is a unit.
- Let $z \in \mathbb{Z}[i]$ be non-zero and not a unit. z is a **prime Gaussian integer** if whenever $z \mid (u \cdot v)$ for $u, v \in \mathbb{Z}[i]$, $z \mid u$ or $z \mid v$.
- $z \in \mathbb{Z}[i]$ is prime iff and only if it is irreducible

Theorem: Division Algorithm For Gaussian Integers

Let $z, w \in \mathbb{Z}[i]$ with $w \neq 0$. Then $\exists q, r \in \mathbb{Z}[i]$ with $z = qw + r$ and $|r| < |w|$.

Definition: GCD For Gaussian Integers

Let $a, b \in \mathbb{Z}[i]$ not both zero. A **greatest common divisor** d of a and b is a Gaussian integer which divides both of them, and any other divisor of a and b divides d .

Lemma: Bézout Lemma For Gaussian Integers

Let $a, b \in \mathbb{Z}[i]$ not both zero. Then d , the greatest common divisor of a and b , exists, and so do $s, t \in \mathbb{Z}[i]$ s.t. $as + bt = d$.

Theorem: Writing Integers As A Sum Of Two Squares

Let $p \equiv 1 \pmod 4$ be a prime. Then $p = a^2 + b^2$ for some $a, b \in \mathbb{Z}$.

- If prime $p \equiv -1 \equiv 3 \pmod 4$, then p is a prime Gaussian integer and $\nexists a, b \in \mathbb{Z}$ s.t. $p = a^2 + b^2$

Let $1 < n, n \in \mathbb{N}$. Suppose $n = 2^c p_1^{\alpha_1} \dots p_k^{\alpha_k} d_1^{d_1} \dots d_t^{d_t}$ is the prime factorisation of n into positive primes, with each $p_i \equiv 1 \pmod 4$ and each $q_i \equiv 3 \pmod 4$ where $c, k, t \geq 0, \in \mathbb{Z}$ and $a_1, \dots, a_k, d_1, \dots, d_t > 0 \in \mathbb{Z}$. Then $\exists a, b \in \mathbb{Z}$ with $n = a^2 + b^2$ iff each d_i is even.

Definition: Associate Integral Domain Elements

Let R be an integral domain and $a, b \in R$. We say a is an **associate** of b if $a = bu$ for u a unit in R .
• If a is an associate of b , b is an associate of a .

Theorem: The FTA For Gaussian Integers

Let $a \in \mathbb{Z}[i]$ be nonzero. Then either a is a unit or a may be expressed as a product of finitely many primes. Moreover, if $a = p_1 \dots p_k = q_1 \dots q_t$ where all p_i, q_j are prime Gaussian integers then the p_i and q_j can be paired such that paired primes are associates.

Example: Checking Irreducibility

Is $z = 4 + i$ an irreducible Gaussian integer?
Yes: z is non-zero and not a unit (since $|z| \neq 1$). To prove it is irreducible we must show that, if $u, v \in \mathbb{Z}[i]$ and $z = u \cdot v$, then u is a unit or v is a unit. Let $z = uv$, then $|z|^2 = |4 + i|^2 = 17 = |uv|^2 = |u|^2 |v|^2$. Since 17 is prime and $|u|^2, |v|^2 \geq 0$, this implies that either $|u|^2 = 1$ or $|v|^2 = 1$, ie that either u or v has absolute value 1 and is thus a unit.

Example: Equality Of Prime Components

Let p be prime and $x, y, z, w \in \mathbb{N}$ s.t. $x^2 + y^2 = p = z^2 + w^2$. Suppose $x \neq z$. Does it follow that $x = w$?
Observe that $(x + iy)(x - iy) = p = (z + iw)(z - iw)$. Note $x + iy$ cannot be a unit as then $x - iy$ would also be a unit and so the modulus of their product would be 1, which is not prime. Thus neither $x + iy$ or $x - iy$ are units, by the same logic neither are $z + iw$ and $z - iw$, so p is not a prime GI (as it is the product of two non-units). By the FTA for Gaussian integers we can write p as $q_1 \dots q_n$ where each q_i is a prime GI, since p is not a prime GI, $n > 1$. Since each q_i is not a unit, $|q_i|^2 \neq 1$ for all i , and note that $|q_i|^2$ divides $p^2 = |p|^2$, so $i = 2$ (dunno how this step works, Agata gave no detail ☺). Now $(x + iy)(x - iy) = q_1 \cdot q_2$, so q_1 divides $x + iy$ or $x - iy$, and $x + iy$ is an associate to either q_1 or q_2 . Similarly $w + iz$ is an associate to either q_1 or q_2 are associates. Therefore, $x + iy$ is an associate to either $z + iw$ or $z - iw$, and it follows that $x + iy \in \{z + iw, -z - iw, w + iz, -w - iz\}$. By assumption $x \neq z$ so $x + iy \neq z + iw$. Since all of x, y, z, w are natural and hence non-negative, $x + iy \neq -z - iw$ and $x + iz \neq w + iz$, so $x + iy = w + iz$ and $x = w$ as required.

Example: Random Gaussian Integer Existence Question

Is it true that for all $n > 0 \in \mathbb{Z}$, there exists z_n, w_n s.t. $(1 + 3i)^n w_n = (2 - i) z_n - n$?
Note that $1 + 3i = (1 + i)(2 + i)$; $2 - i = 1 + i + 2 + i$ are all pairwise coprime GIs. Thus $(1 + 3i)^n$ and $(2 - i)^n$ are coprime GIs, as any common divisor of them can be written as a product of prime GIs and a unit. By the Bézout Lemma there exist GIs p_n, q_n s.t. $(1 + 3i)^n p_n + (2 - i)^n q_n = 1$, so we may take $w_n = -p_n \cdot n$ and $z_n = q_n \cdot n$, thus such z_n, w_n always exist.

Example: Sum Of Squared Integers

Does there exist $a, b \in \mathbb{Z}$ s.t. $a^2 + b^2 = 77$?
Note $77 = 7^1 \cdot 11^1$. Since 7 and 11 are prime GIs, appear with odd powers in 77, and are congruent to 3 modulo 4, by lecture theory no such a, b exist.

Example: Sum Of Squared Gaussian Integers

Does there exist $w, z \in \mathbb{Z}[i]$ s.t. $w^2 + z^2 = 61$?
Yes: Observe $w^2 + z^2 = (w + zi)(w - zi)$. Thus if we can choose w, z s.t. $w + zi = 61$ and $w - zi = 1$ we will satisfy our equation. Rearrange to see that $2w = (w + iz) + (w - iz) = 61 + 1 = 62$ so $w = 62/2 = 31$; and $2iz = (w + iz) - (w - iz) = 61 - 1 = 60$ so $z = 60/2i = -30i$ satisfy our equations.

Ex.: Showing An Integer Is Not A Sum Of Squared GIs

Show that $q \in \mathbb{Z}$ of the form $q = 4k + 3$ for some $k \in \mathbb{Z}$ is not a sum of two squares of integers. Note that for all $x \in \mathbb{Z}$, $x^2 \equiv 1 \pmod 4$ or $x^2 \equiv 0 \pmod 4$. Thus $x^2 + y^2$ has remainder 0, 1, or 2 modulo 4, so cannot be congruent to 3 modulo 4, but q is! Thus no such x, y exist.

Ex.: A Prime Integer Of Form $4k + 1$ Is Not A Prime GI

Let $p > 2$ be a prime number of form $4k + 1$ for some $k \in \mathbb{Z}$. Show that p is not a prime GI. A GI is prime iff it is irreducible, so we prove p is not irreducible. By lecture theorem we know since $p = 4k + 1$ for some $k \in \mathbb{Z}$, that $p = x^2 + y^2$ for some $x, y \in \mathbb{Z}$. Since p is prime, $x, y \neq 0$. Note also that $p = (x + iy)(x - iy)$, so it suffices to show that neither $x + iy$ or $x - iy$ is a unit. Suppose one of them was a unit. Then $\exists v \in \mathbb{Z}[i], v = a + bi$, s.t. either $(x + iy)v = 1$ or $(x - iy)v = 1$. Then either $|x + iy||v| = 1$ or $|x - iy||v| = 1$, in either case implying that $\sqrt{x^2 + y^2} \cdot \sqrt{a^2 + b^2} = 1$. However, since $x, y \neq 0$, $x^2 + y^2 > 1$ and since $a, b \in \mathbb{Z}$, $|v| \geq 1$, thus we have a contradiction.

Example: A Prime Integer Of Form $4k + 3$ Is A Prime GI

Let $p > 2$ be a prime number of form $4k + 3$ for some $k \in \mathbb{Z}$. Show that p is a prime GI. A GI is prime iff it is irreducible, so we prove p is irreducible. Suppose for contradiction $p = (x + iy)(a + ib)$ for some $x, y, a, b \in \mathbb{Z}$ s.t. neither of $x + iy$ and $a + ib$ are units. Then $|p| = |x + iy||a + ib|$, and $|p|^2 = p^2 = (x^2 + y^2)(a^2 + b^2)$. By the second example before this one, $p \neq x^2 + y^2$ and $p \neq a^2 + b^2$. Thus it cannot be true that $|x + iy|^2 = 1$ or that $|a + ib|^2 = 1$, therefore neither of $x + iy$ and $a + ib$ have modulus 1 and thus neither can be a unit. Thus p is an irreducible GI and therefore a prime GI.

Ex.: Prime Integer $q \equiv 3 \pmod 4$ With $q \mid (x^2 + y^2)$ Divides x And

Let $q \equiv 3 \pmod 4$ be prime and suppose that $q \mid (x^2 + y^2)$ for some $x, y \in \mathbb{Z}$. Prove that $q \mid x$ and $q \mid y$.
By the previous example q is a prime and irreducible GI. Since $x^2 + y^2 = (x + iy)(x - iy)$, $q \mid (x + iy)(x - iy)$, so q divides one of $x + iy$ and $x - iy$. If $q \mid (x + iy)$ then $q \mid (a + ib) = (x + iy)$ for some $a + ib \in \mathbb{Z}[i]$, thus $qa = x$ and $qb = y$, ie q divides x and y . Similarly if $q \mid (x - iy)$ then $q \mid (a + ib) = (x - iy)$ for some $a + ib \in \mathbb{Z}[i]$, so $qa = x$ and $q(-b) = y$, ie q divides x and y .